PINOUT & CONNECTION GUIDE · SKU PIA24028P

Module Pins, I²C, Crowtail Ports & Electrodes

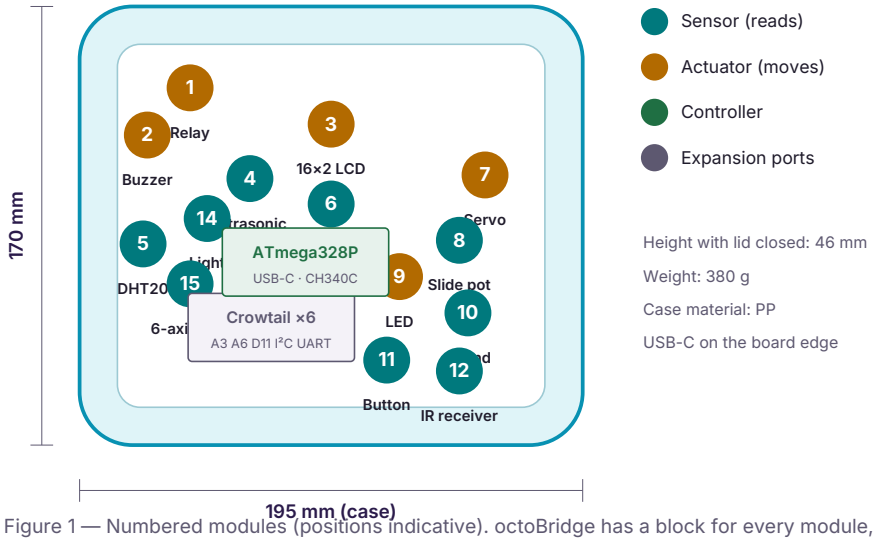
Where every module on the starter board connects, the I²C addresses, the six Crowtail expansion ports, the buttons and LEDs, and the Octopus 16 electrode array and XIAO orientation.



16	4	6	18	115200
Modules on the board	I²C devices	Crowtail ports	Pogo pins	Baud over USB

01 STARTER BOARD MODULES

STARTER BOARD IN ITS CASE · TOP VIEW · POSITIONS INDICATIVE



- Sensor (reads)
- Actuator (moves)
- Controller
- Expansion ports

Height with lid closed: 46 mm
Weight: 380 g
Case material: PP
USB-C on the board edge

Figure 1 — Numbered modules (positions indicative). octoBridge has a block for every module, so pin numbers are only needed for Arduino code.

02 MODULE PIN TABLE

Module	Board pin	MCU pin	In octoBridge
Red LED	D10	PB2	light D10: on / off / flash
Second light	D8 *	PB0	light D8: on / off
Buzzer	D3	PD3 (PWM)	buzzer: on / off
Relay	D4	PD4	relay: on / off — never mains
Servo	D9	PB1 (PWM)	servo 0–180°
IR receiver	D2	PD2 (INT0)	remote key code
Button	D7	PD7	pressed / released
Ultrasonic	D6 trig · D5 echo	PD6 · PD5	distance, cm
Slide potentiometer	A0	PC0	knob, raw 0–1023
Sound sensor	A1	PC1	raw 0–1023
PIR motion	A2	PC2	motion / no motion

Module	Board pin	MCU pin	In octoBridge
Soil moisture (Crowtail)	A3	PC3	raw 0–1023
Light BH1750	I²C 0x5C	PC4 SDA · PC5 SCL	lux
Temperature & humidity DHT20	I²C 0x38	PC4 · PC5	°C, %
6-axis IMU (LSM6DS3 or MPU6050)	I²C 0x6B / 0x68	PC4 · PC5	tilt (no heading)
16×2 LCD (via I²C expander)	I²C 0x21	PC4 · PC5	2 lines × 16 characters

* Light D8 is driven by the OctoBio firmware; Elecrow's datasheet lists D8 (PB0) as reserved and only one LED (D10). Raw counts 0–1023 are not calibrated units.

03 I²C BUS

Device	Address
Light sensor BH1750	0x5C
DHT20 temperature & humidity	0x38
6-axis IMU: LSM6DS3 / MPU6050	0x6B / 0x68
16×2 LCD (I²C expander)	0x21

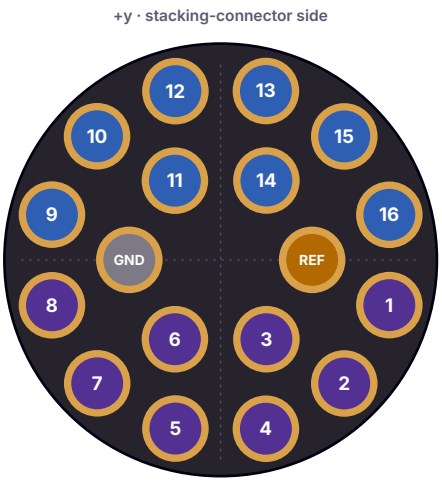
All four devices share SDA (A4, PC4) and SCL (A5, PC5). Which IMU is fitted depends on the board version; the firmware handles both. The two I²C Crowtail ports are on the same bus, so avoid add-on sensors at these addresses.

04 CROWTAIL PORTS, BUTTONS AND LEDS

Port / part	Pins	Use
A interface ×2	A3 (PC3) · A6 (ADC6)	Analog input — soil moisture sensor on A3
D interface	D11 (PB3)	Digital I/O
I²C ×2	A4/SDA · A5/SCL	I²C add-ons (G, 5V, SDA, SCL)
UART	D0/RX · D1/TX	Shared with USB — leave free while octoBridge is connected
RESET button	PC6 (/RESET)	Restarts the board
Button	D7 (PD7)	User button
LEDs	PWR · RX · TX	Red power LED; blue LEDs flash with serial traffic

Crowtail pins are 5 V, 20 mA each, with G, 5V and signal on each connector.

05 OCTOPUS 16 ELECTRODE ARRAY



Pins	Count	Purpose
Signal 1–8	8	ADS131M08 #1
Signal 9–16	8	ADS131M08 #2
REF	1	Common reference
GND	1	Ground

All 18 pins sit in one ~2 × 2 cm cluster, so the whole montage rests on one spot of the scalp. For the OctoBio lessons, place it over the back of the head (Oz / Pz).

WARNING XIAO orientation.

A XIAO mounted 180° the wrong way still flashes and shows up on Bluetooth but sends no EEG (LED double-blinks). Match the 5V / GND / 3V3 labels.

Figure 2 — Pin-side view. Coordinates and jumpers: Octopus 16 set, OC16-PIN-004.

06 RELATED DOCUMENTS

OB-DS-001 Product Datasheet	**OB-MD-002** Mechanical & Layout Drawing	**OB-PCB-003** Hardware & Manufacturing Specification
OB-QSG-005 Quick Start & Safety Guide	**OB-SW-006** Software & Protocol Guide	